



Data quality for big data

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Big data, big problems?

- Big data -> automation
- Example:

Ground Motion Visualization (GMV) code

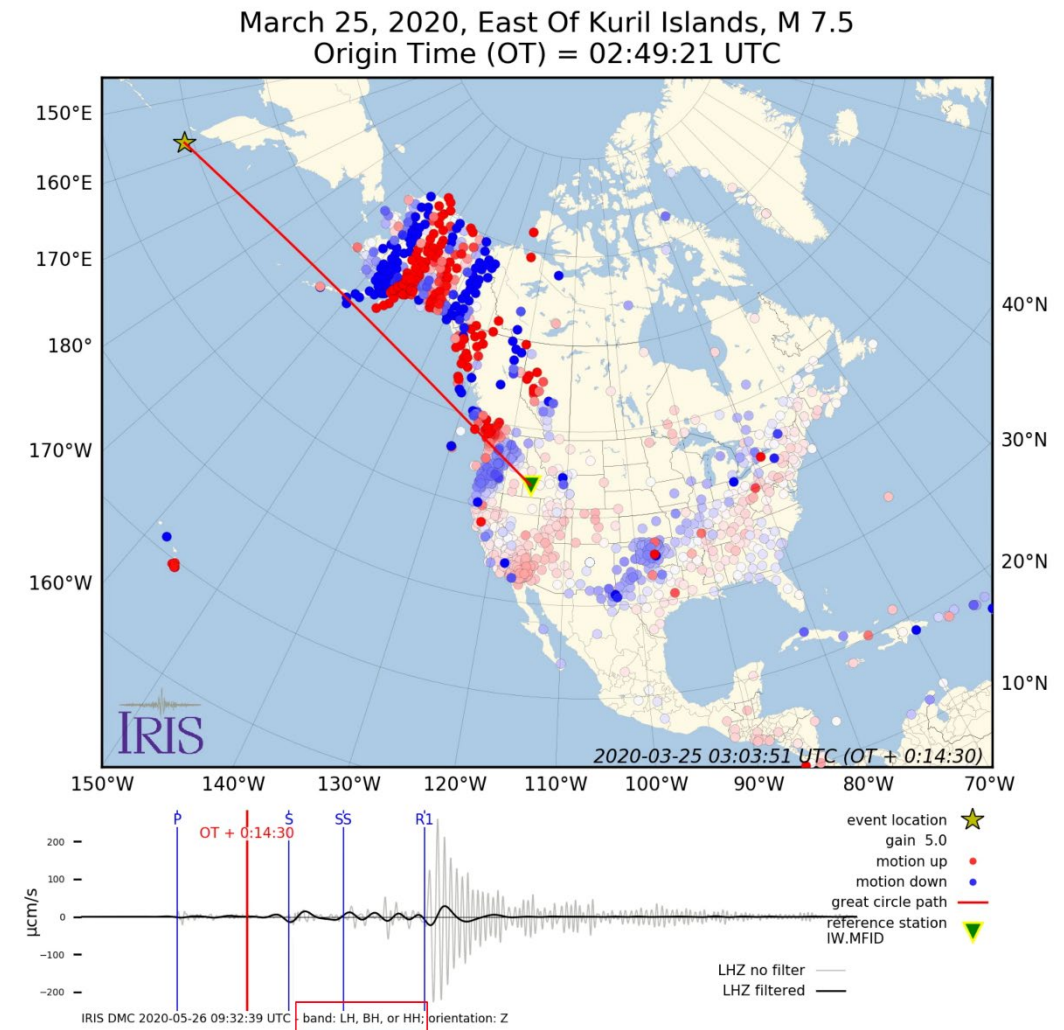
<https://github.com/iris-edu/gmv>

<http://ds.iris.edu/ds/products/gmv/>

- fully automated, all in one

Trabant et al., 2012

- Quality control of input data is a must
-> automation?



Quality control and State of Health



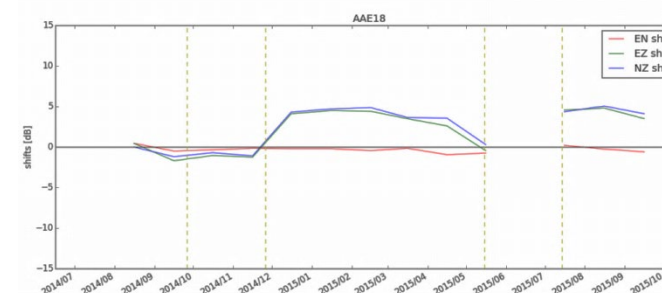
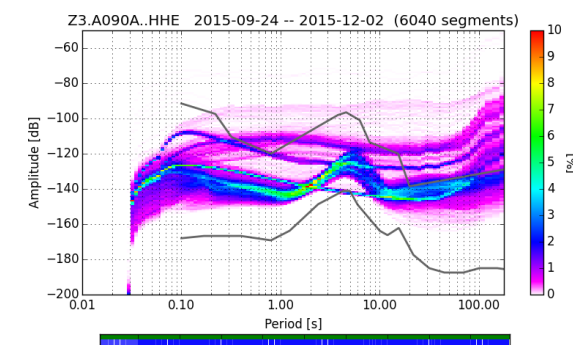
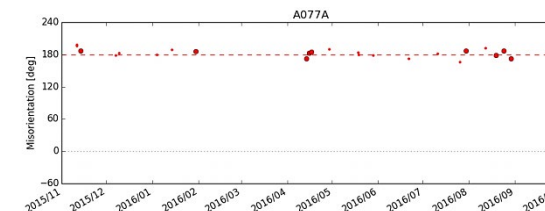
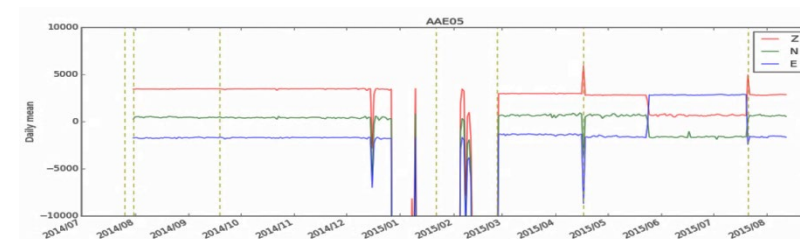
hardware methods



Data availability
Gaps, overlaps in data
Reversed or interchanged components
Metadata
Sensor orientation
Signal clipping
Drift of sensor mass position
Amplitude gain imperfection
Noise in signals
Glitches in signal
Timing issues

...

software methods



Station metadata - station siting, instrumentation

• network code	Y8	}	must be same as in mseed	• sensor orientation
• station code	AVR			- azimuth
• location code	00			- dip
• channel codes	HHZ, HHN, HHE			• responses
• sample rating	100 [Hz]			- datalogger
				- sensor
• latitude	43.11949 [deg]			
• longitude	27.66911 [deg]			
• elevation	292 [m]			
• time epochs	startDate, endDate			... and more

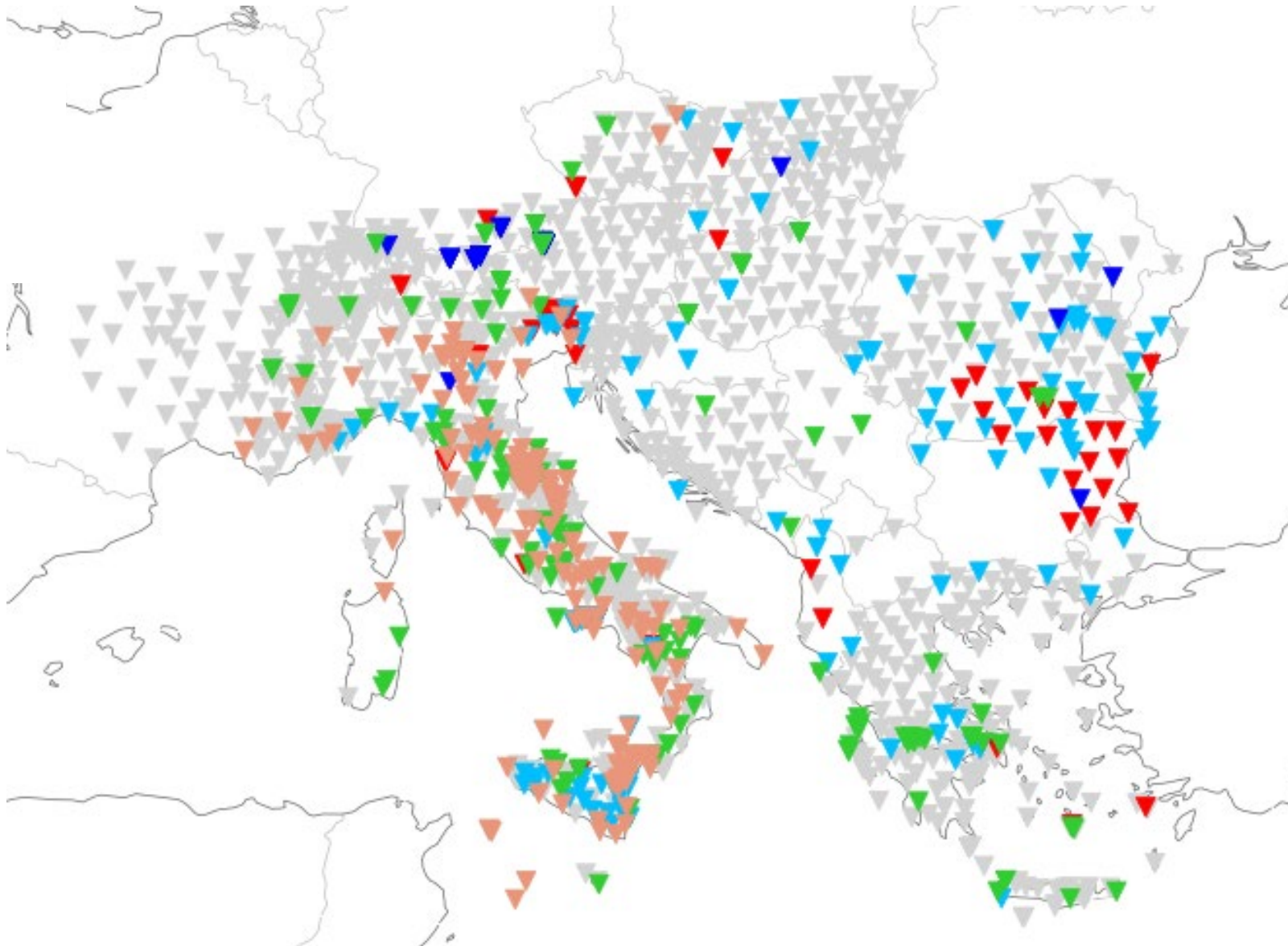
Metadata file formats

- Dataless SEED (binary)
https://ds.iris.edu/ds/nodes/dmc/data/formats/#dataless_seed
- RESP format (ASCII)
 - ASCII representation of Dataless SEED with only instrument response information<https://ds.iris.edu/ds/nodes/dmc/data/formats/resp/>
- **StationXML** (ASCII)
 - a modern alternative to Dataless SEED<http://www.fdsn.org/xml/station/>

All standards promoted by the International Federation of Digital Seismograph Networks (FDSN;
<https://www.fdsn.org/>)

Check of _ADRIAARRAY metadata

Metadata downloaded
from EIDA **29.3.2023**



▴ metadata warnings
(15.3 %)

▴ errors in metadata
(3.6 %)

▴ wrong band code
(8,4 %)

▴ different corner periods
in metadata and the list
(8.8 %)

▴ switched broad-band
and short-period
(1.0 %)

total: 37 %

Sensor mis-orientation

How to measure:

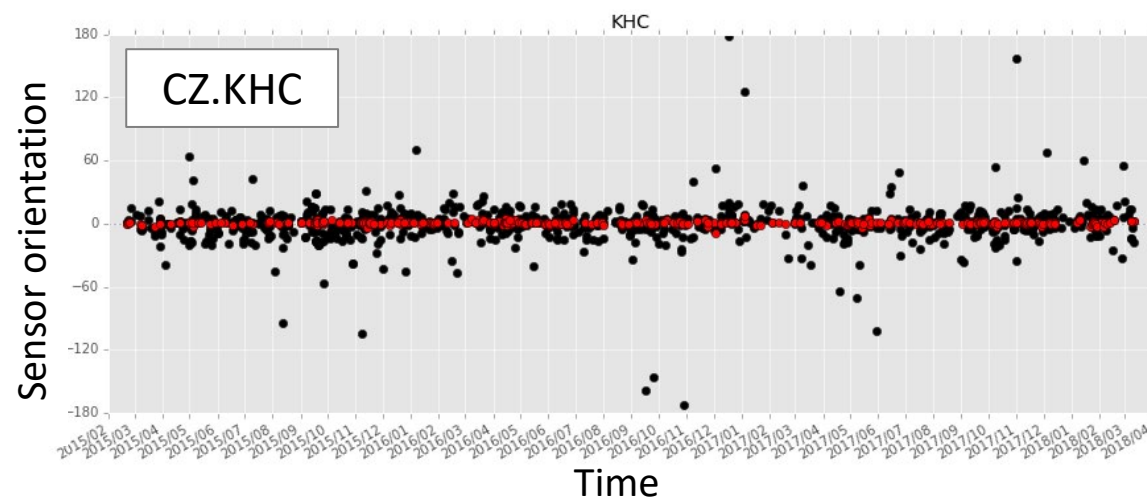
Gyrocompass: very accurate, not very frequent usage (if ever)

Polarization of Rayleigh waves (single station method): not so accurate, better detection of temporal changes (see black dots)

Improved Rayleigh wave method (multi station method): corrected for dynamic back-azimuths, best precision 1-2° (see red dots)

How to correct:

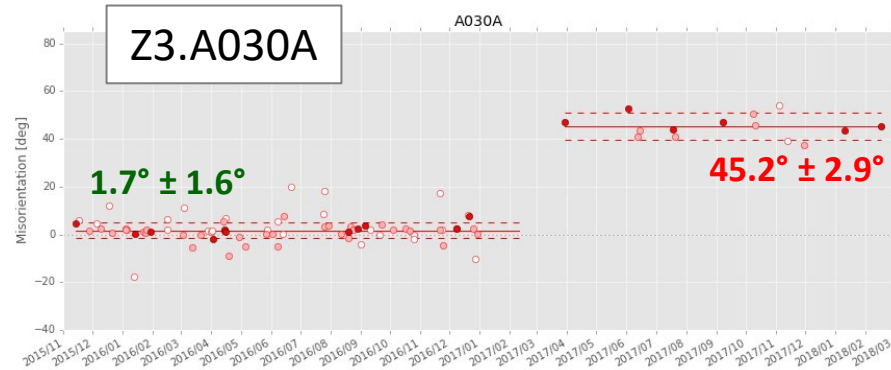
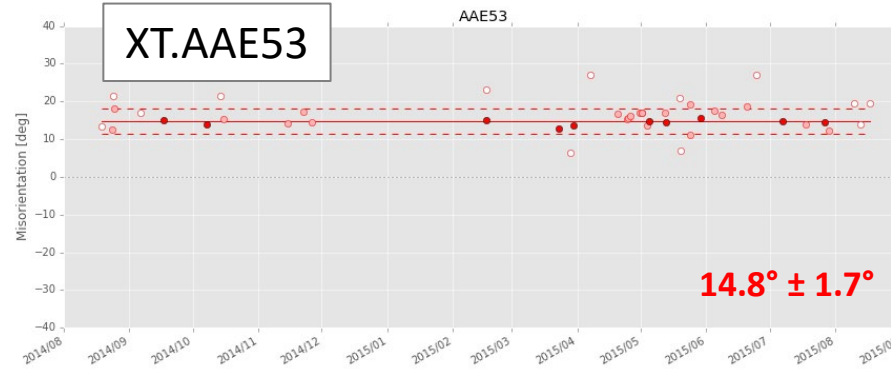
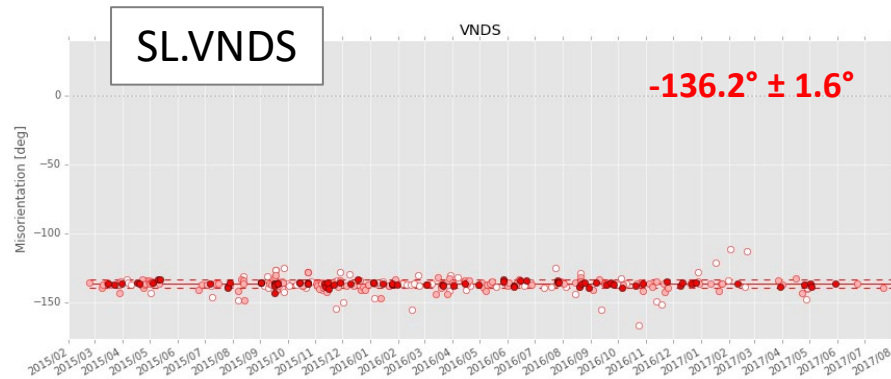
Change values in station metadata (easy!)



Comparison of sensor orientations re-measured by a gyrocompass and computed by the RW polarization method.	Gyrocompass		RW polarization	
	AAE27	8.1°	10.7°	± 3.3°
	AAE28	7.5°	8.5°	± 2.1°
	AAE29	13.2°	13.7°	± 1.3°
	AAE30	-3.6°	-4.0°	± 2.2° (+89°)
	AAE31	6.5°	7.7°	± 2.5°
	AAE32	-0.3°	3.2°	± 2.4°
	AAE33	8.1°	8.4°	± 1.4°
	AAE34	5.4°	6.0°	± 1.5°

Sensor mis-orientations

Sensor orientation



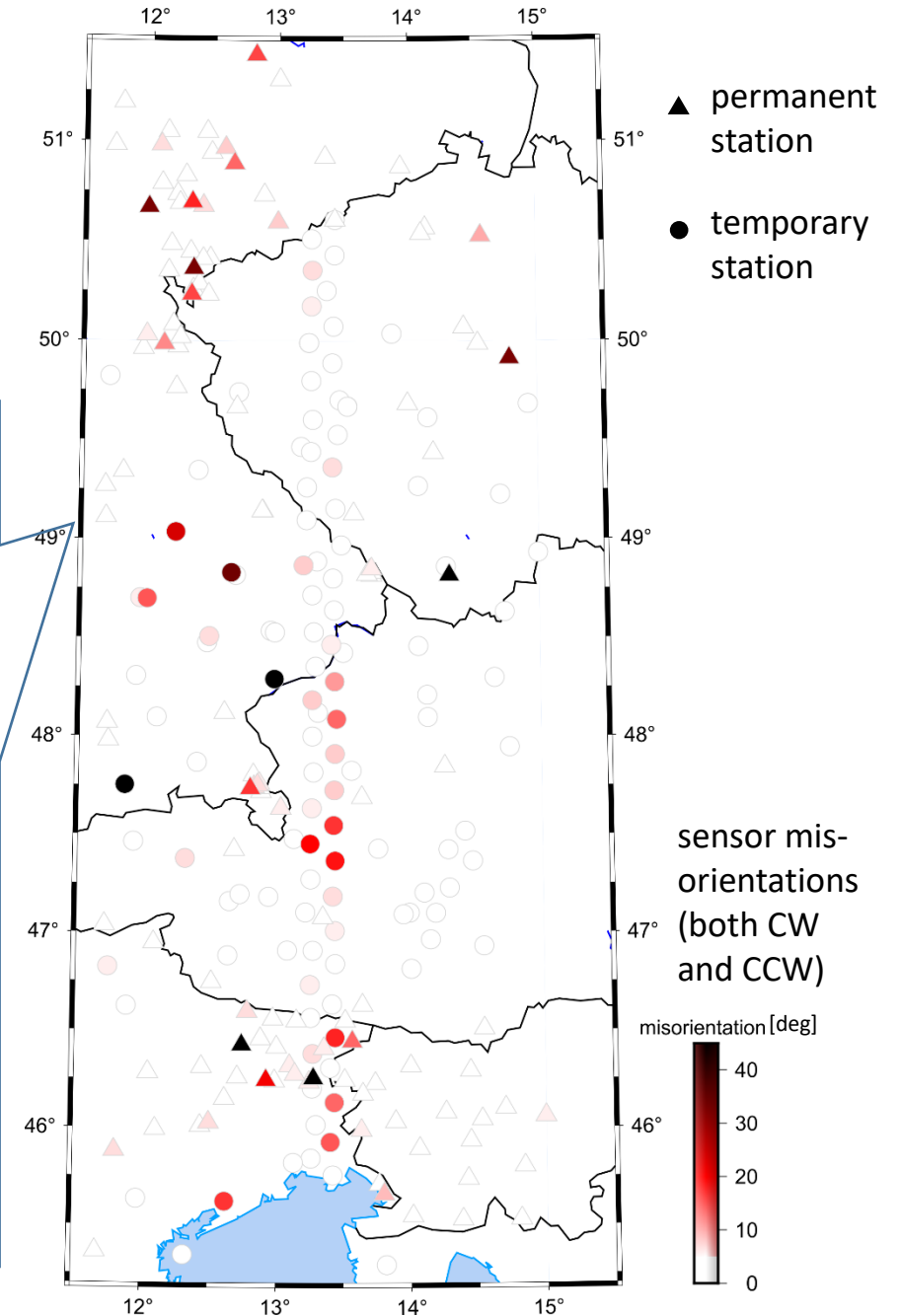
2014-2018: Rayleigh wave method (multi station)

2014-2023: SKS polarization method (single station)

Deviations of sensor orientations from the official orientations given in metadata (from EIDA, 2023-06-05).

Number of stations with sensor mis-orientations:

$\geq 5^\circ$	77/250	31 %
$\geq 10^\circ$	29/250	12 %
$\geq 20^\circ$	12/250	5 %
$\geq 30^\circ$	9/250	4 %
$\geq 45^\circ$	5/250	2 %
$\geq 90^\circ$	3/250	1 %



Daily means – fast and powerful tool for QC

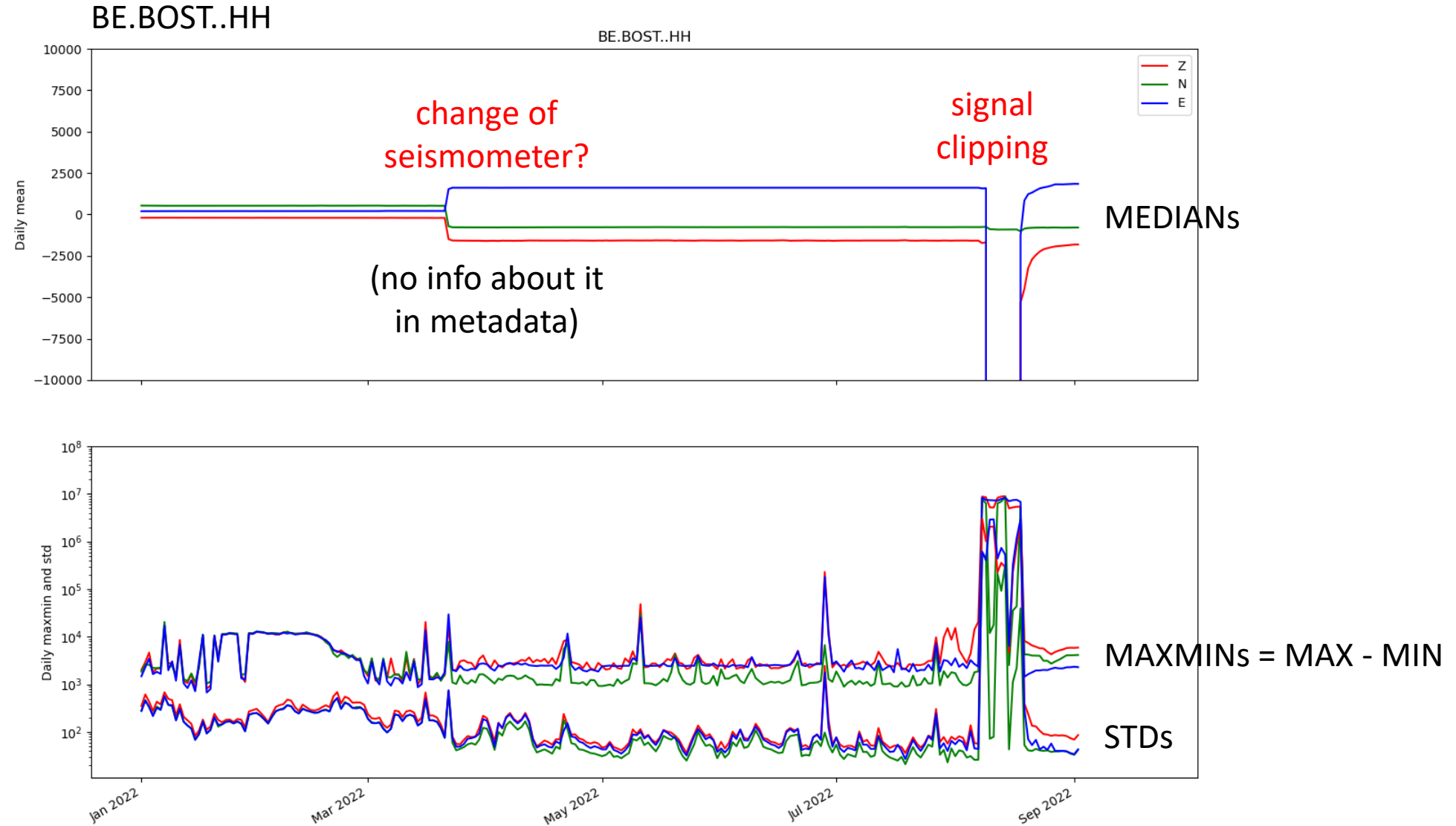
Probabilistic power spectral density (PPSD)

- time-frequency representation of the signal
- not included in EIDA WFCatalog (yet)
- need to download continuous data
- long computation of PPSD
- depends on metadata (changes/updates!)

Metrics in WFCatalog

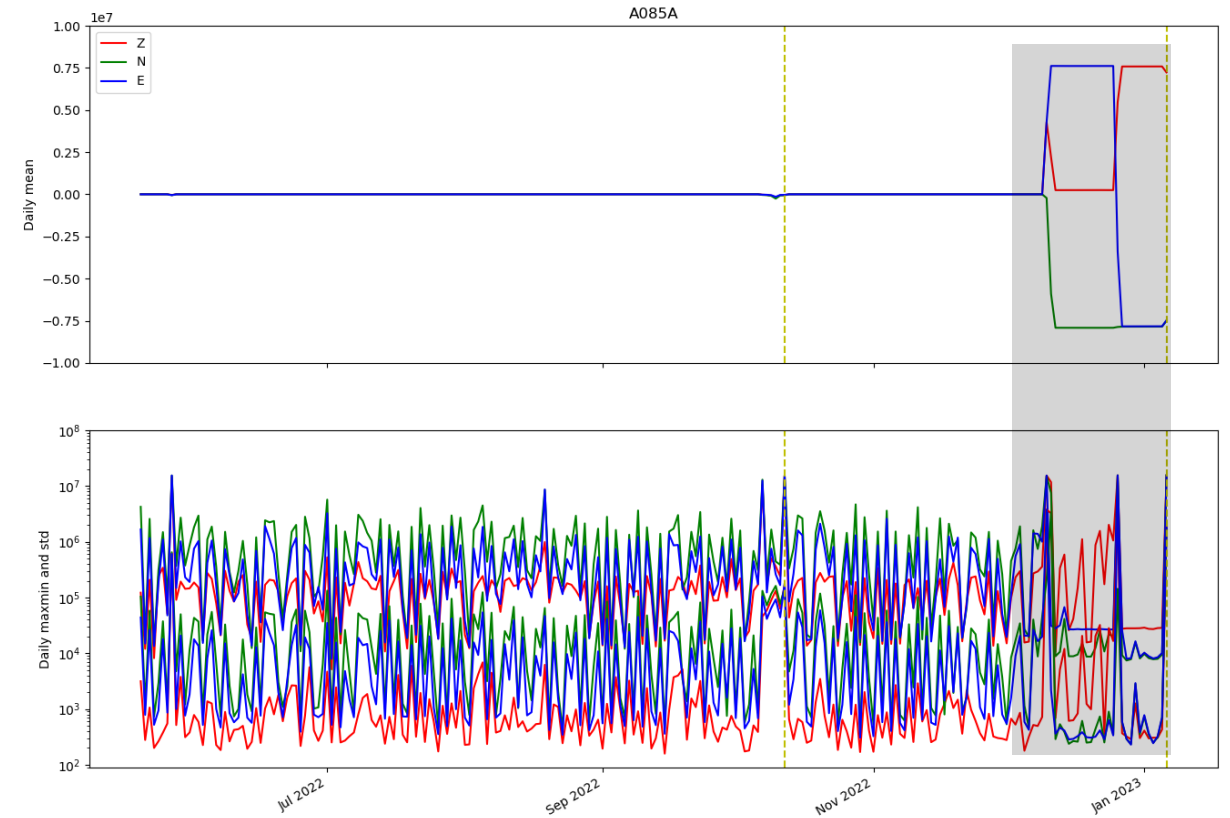
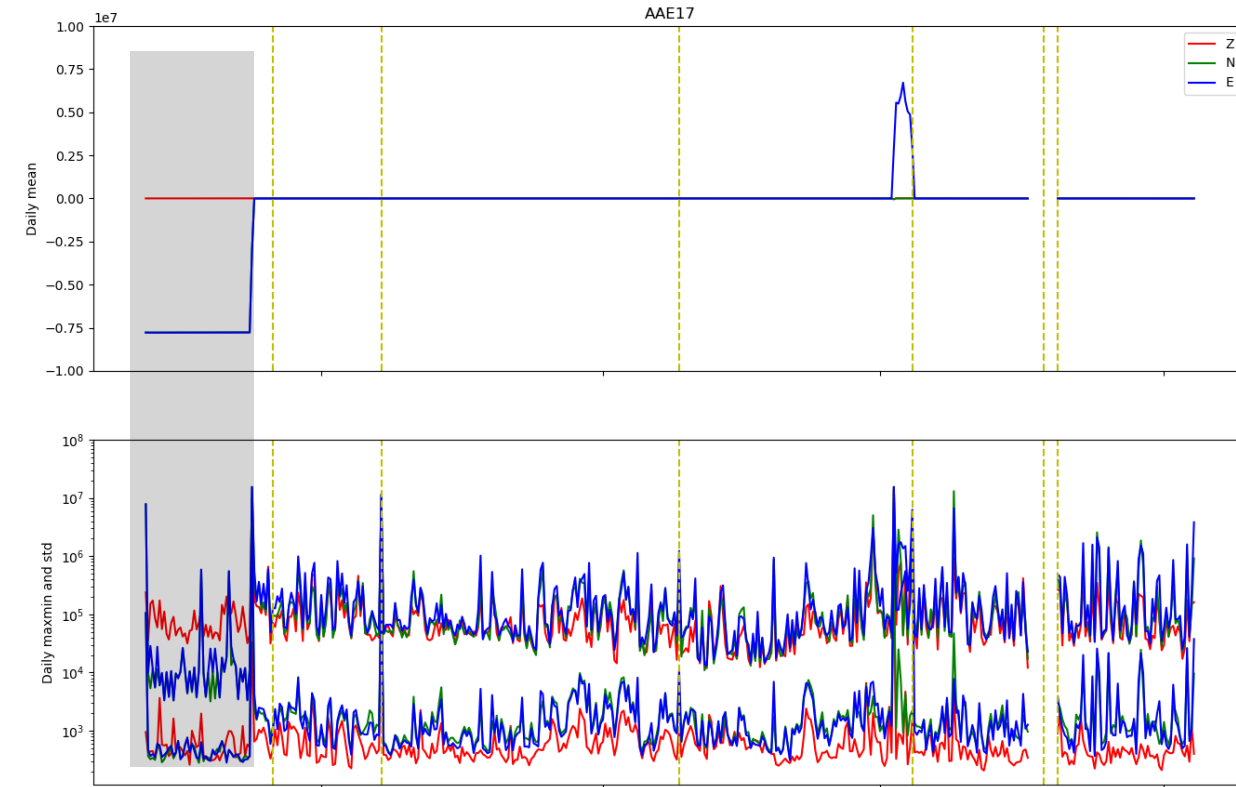
- no frequency analysis
- easy and fast to download from EIDA
- doesn't depend on metadata
- metrics:
 - daily median
 - daily std
 - daily max, min
 - ...

Change of seismometer, signal clipping



Saturated signal

Saturated signal



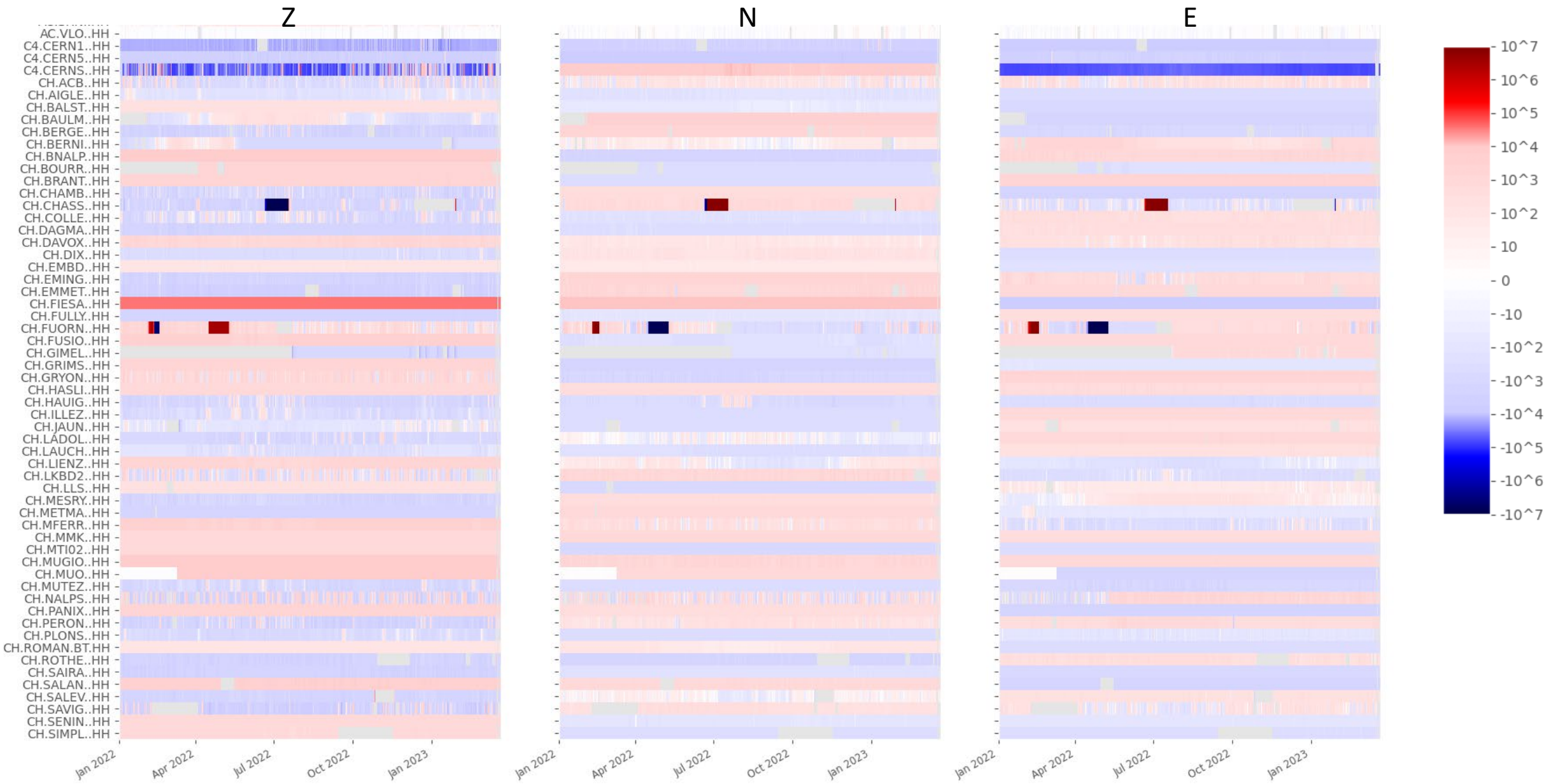
What is signal maximum?

24 bit ADC = 8.4×10^6
32 bit ADC = 2×10^9

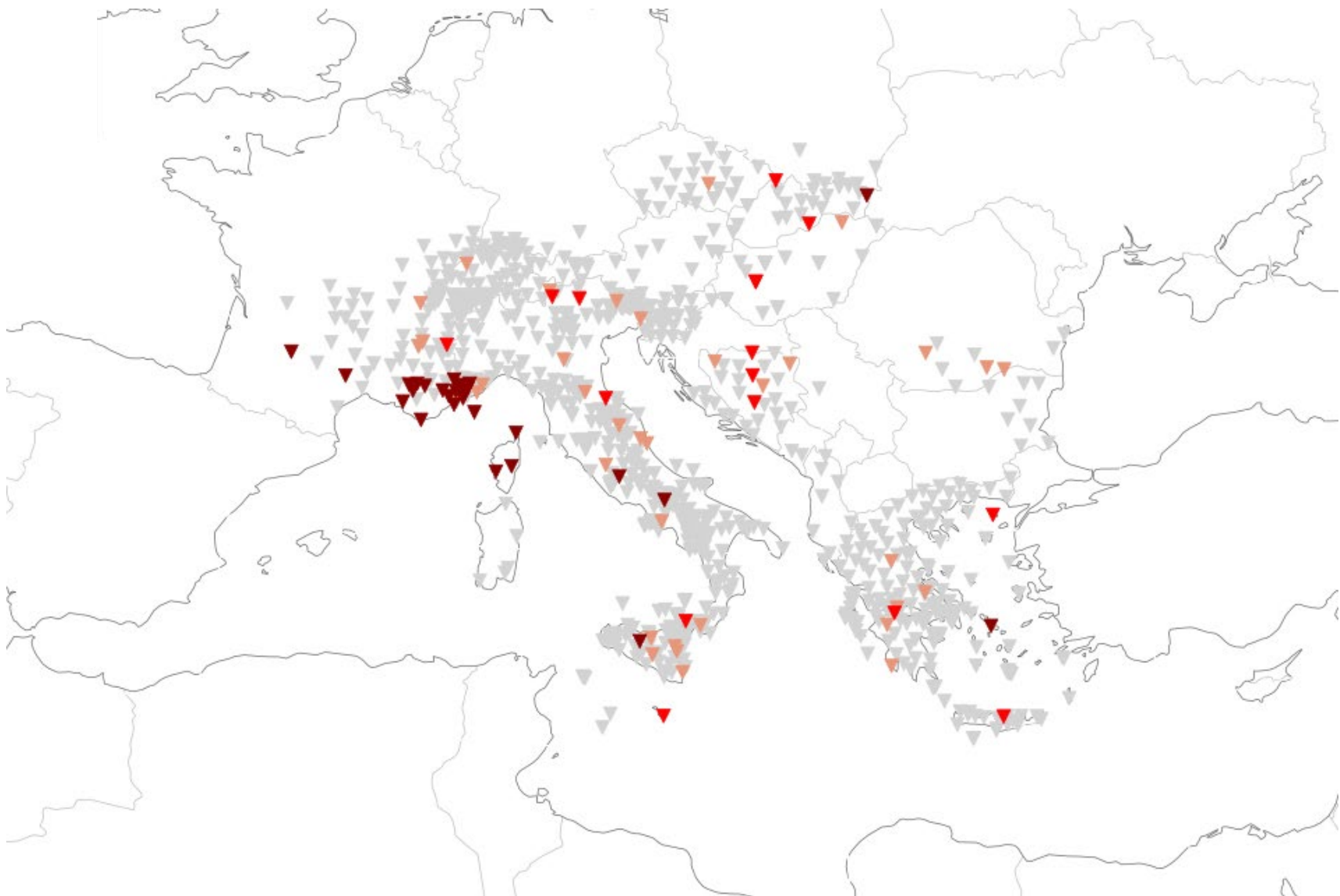
Gray areas:

saturated offsets → saturated amplitudes

Saturation of signal daily medians/offsets



Saturated medians - map



Tested period:
2022-01-01 – 2023-03-21
445 days

Saturation condition:
 $\text{abs}(\text{median}) > 1\text{e}6$

- ▲ all stations (861)
- ▲ 10+ days: 79/861 9 %
- ▲ 100+ days: 43/861 5 %
- ▲ 300+ days: 28/861 3 %

Maximal amplitudes from a seismic event

Amplitude gains

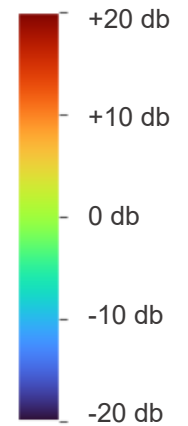
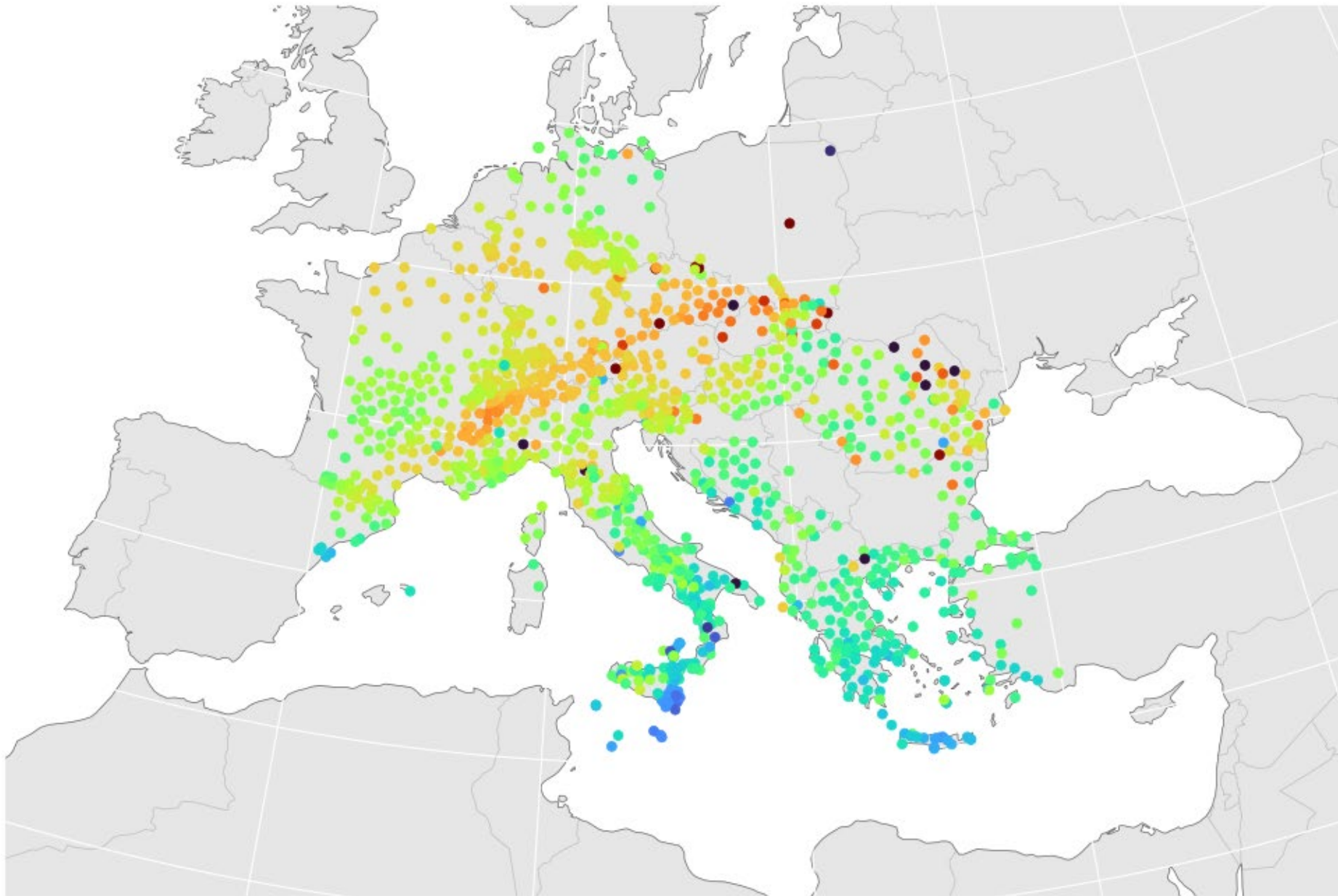
Slovakia 5.0

2023-10-09T18:23

N49.057°, E21.717°

depth 7.3 km

Z component
filtered 20-60 s



Maximal amplitudes from a seismic event, outliers corrected

Amplitude gains

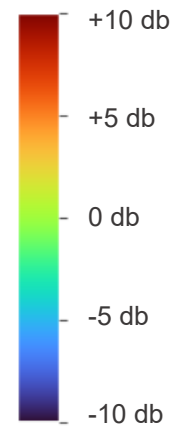
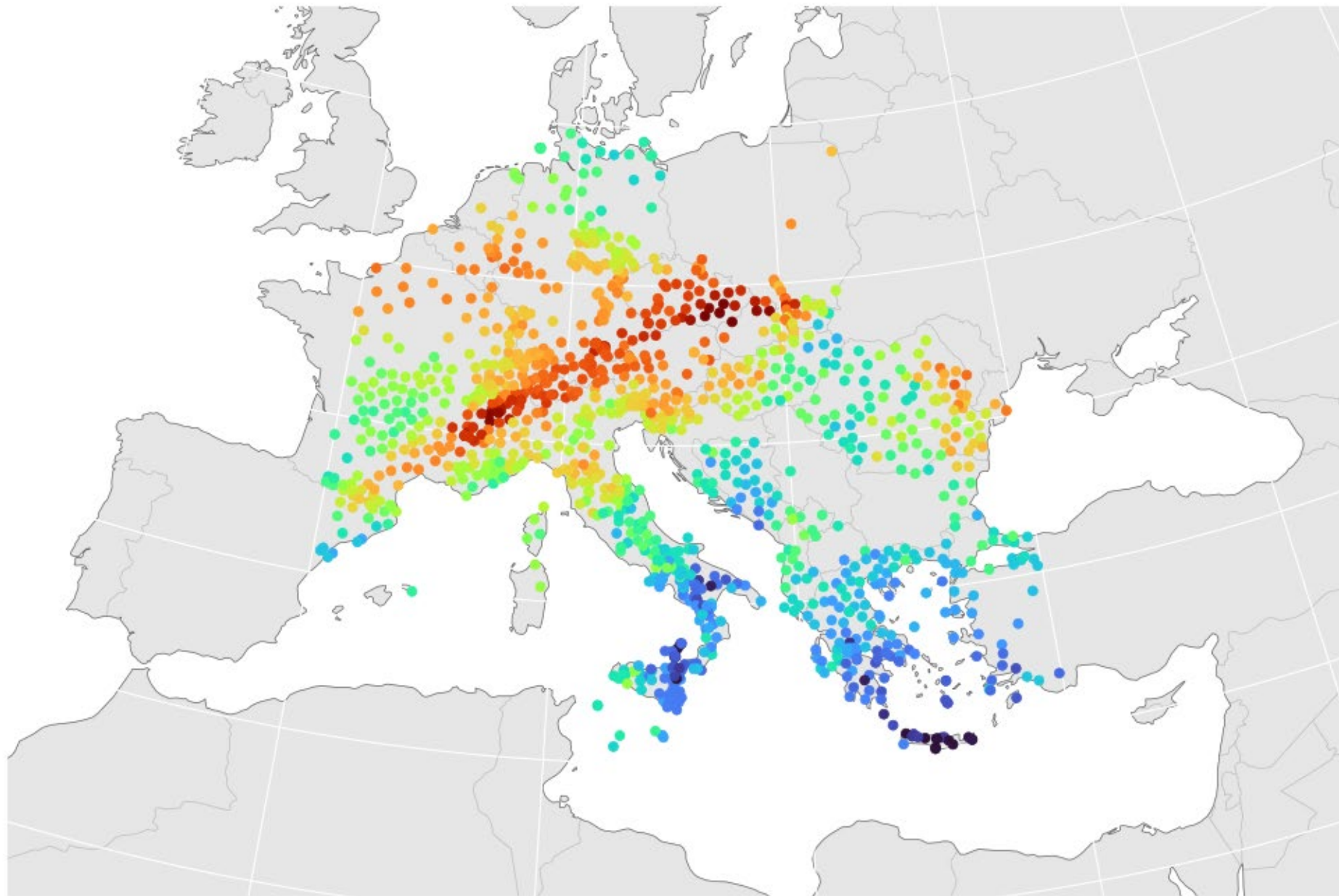
Slovakia 5.0

2023-10-09T18:23

N49.057°, E21.717°

depth 7.3 km

Z component
filtered 20-60 s



Maximal amplitudes from a seismic event

Amplitude gains

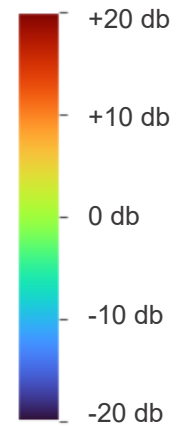
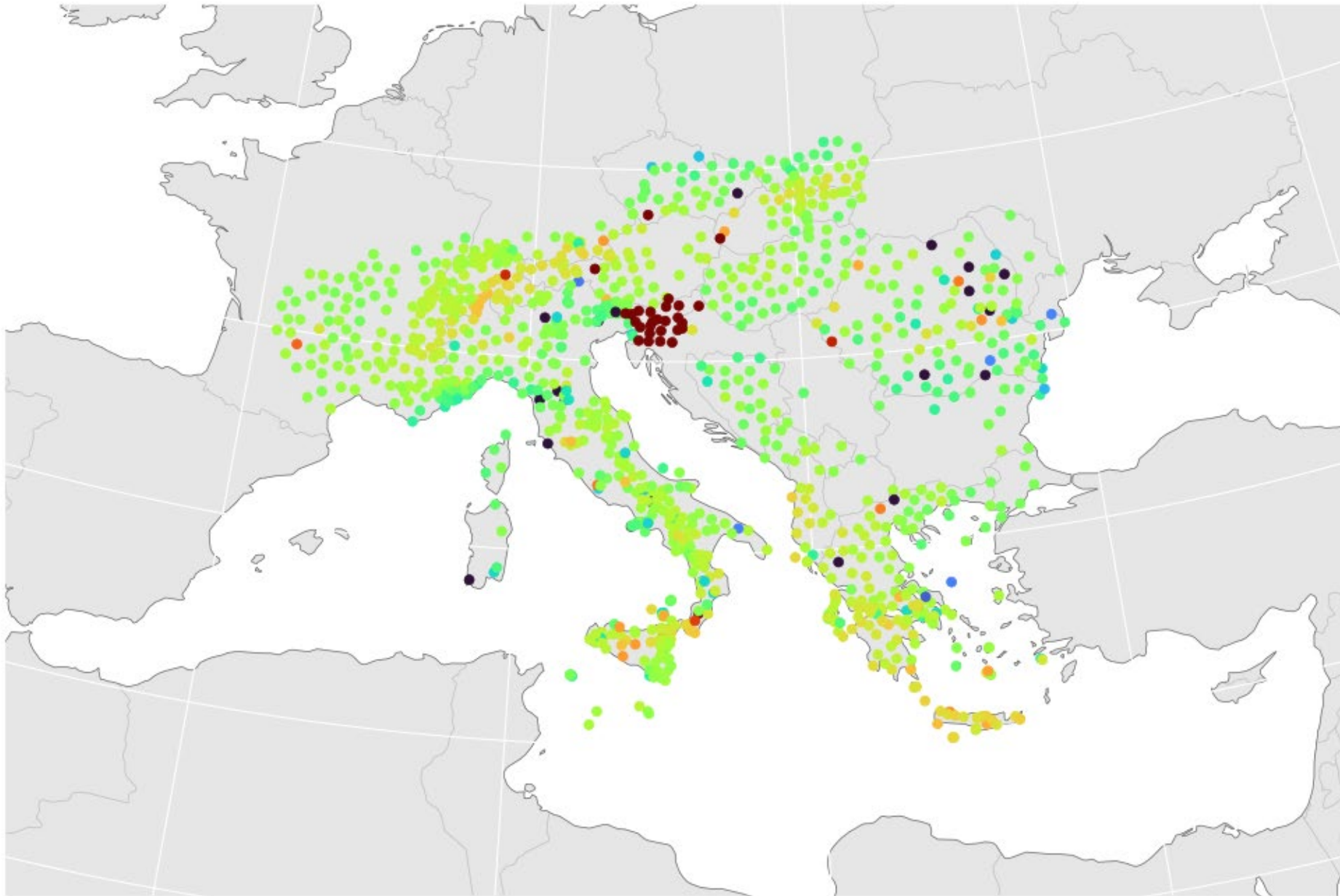
Morocco 6.8

2023-09-08T22:11

N31.055°, W8.389°

depth 19.0 km

Z component
filtered 20-60 s



Maximal amplitudes from a seismic event, outliers corrected

Amplitude gains

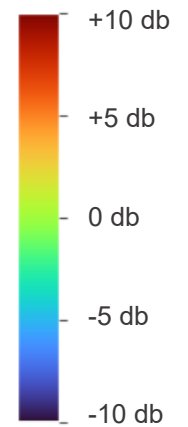
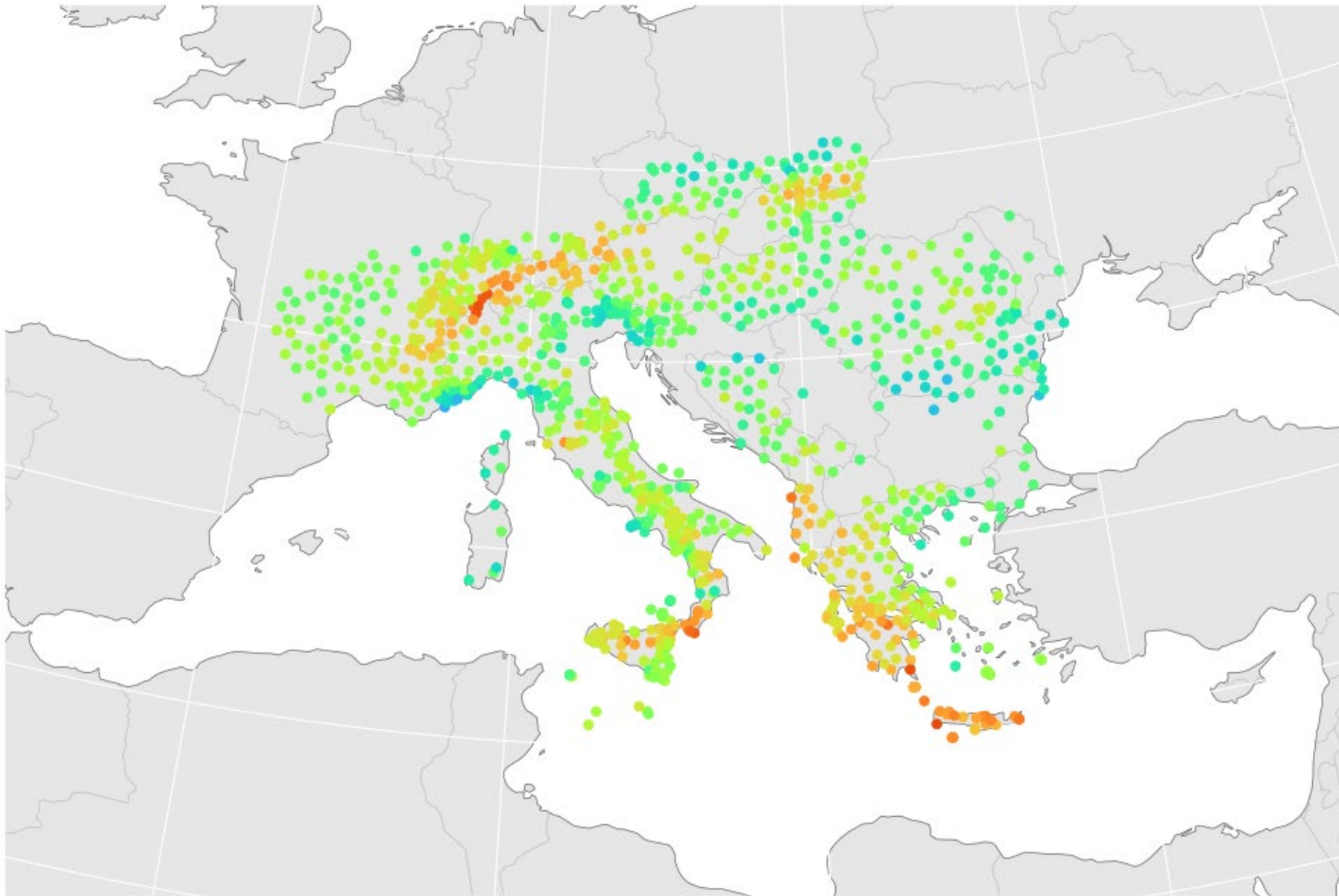
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depth 19.0 km

Z component
filtered 20-60 s

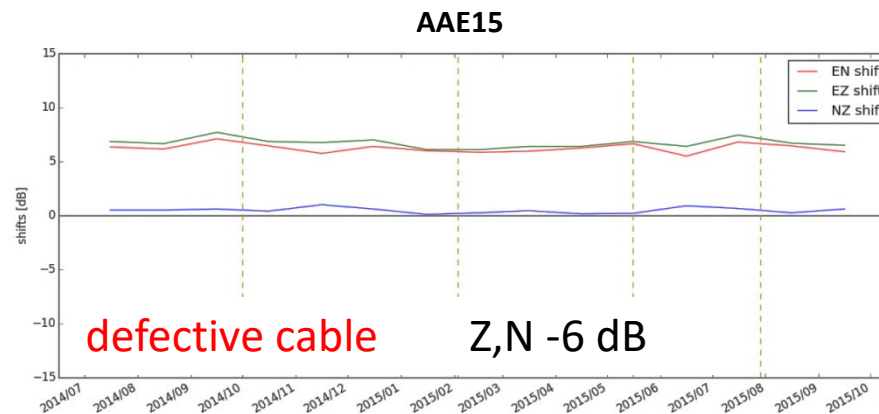
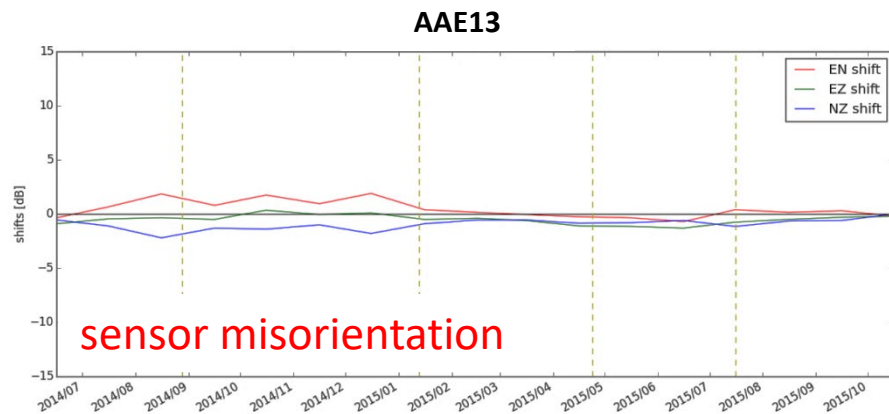
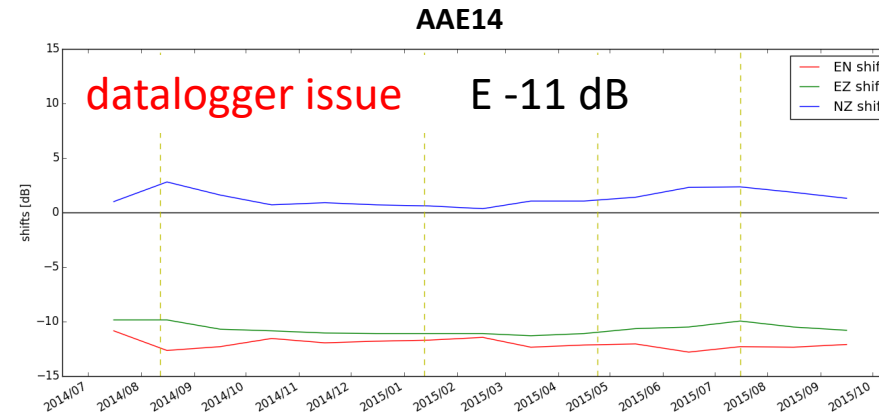
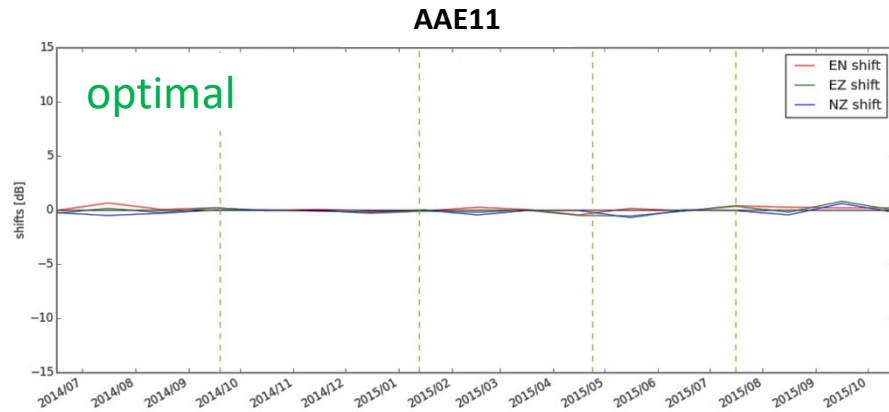
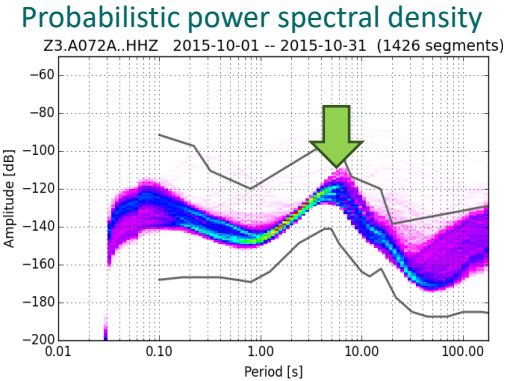


Detection of amplitude gain issues by the PPSD method

It compares ratios of PPSDs between the three components in a range of 4-8 seconds (where the secondary microseisms are substantially larger than noise from local sources).

A global and local trends of the ratios are removed.

Double check - stations with amplitude misgains are tested by special control devices for seismometers (CMG, STS) and GAIA DAs (developed in IG CAS).



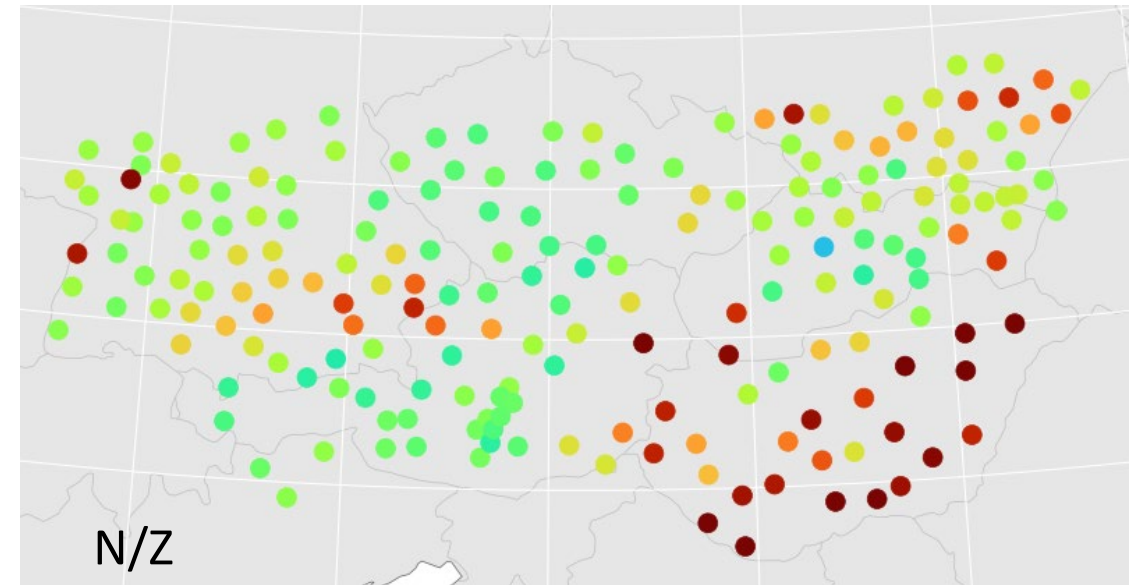
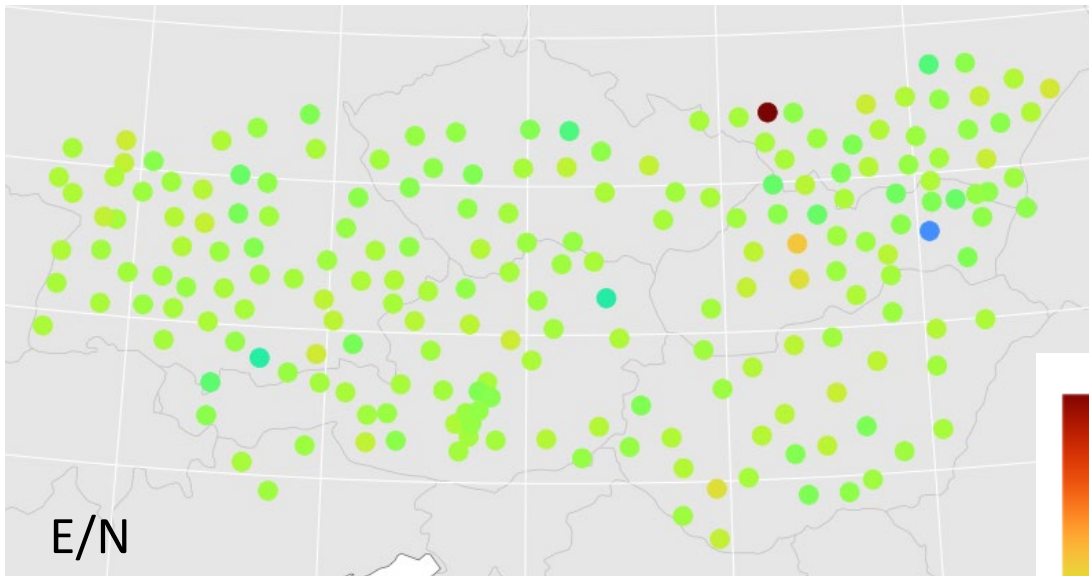
Control and calibration units



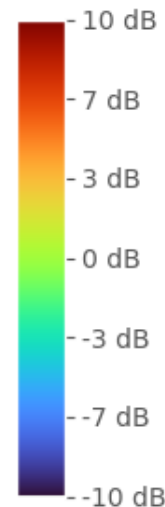
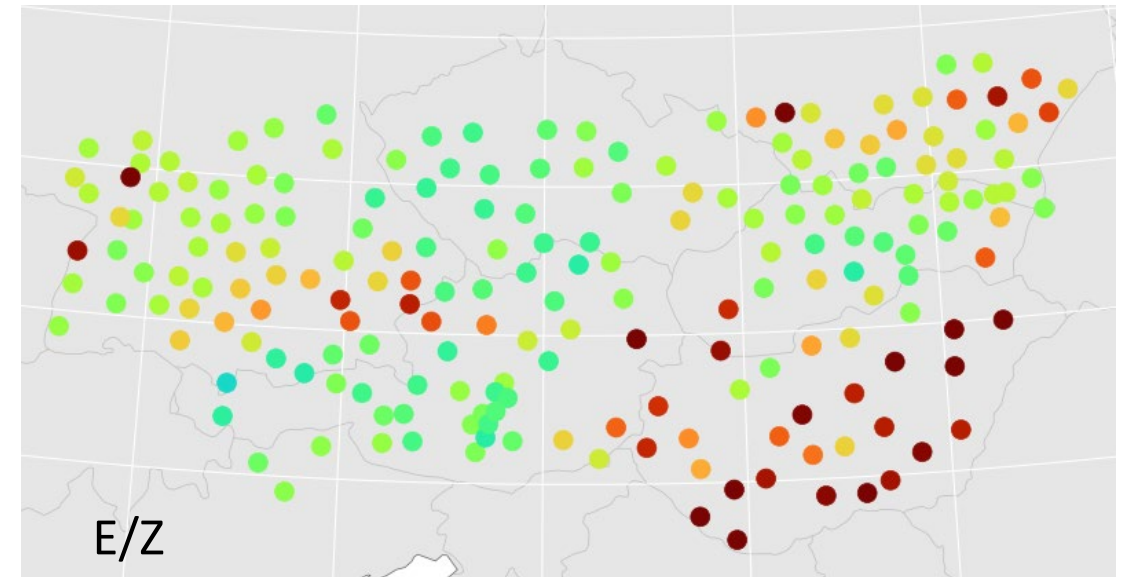
PACASE - amplitude gain issues

- average PPSD ratios for $T = 5$ s
- global trend subtracted

outliers

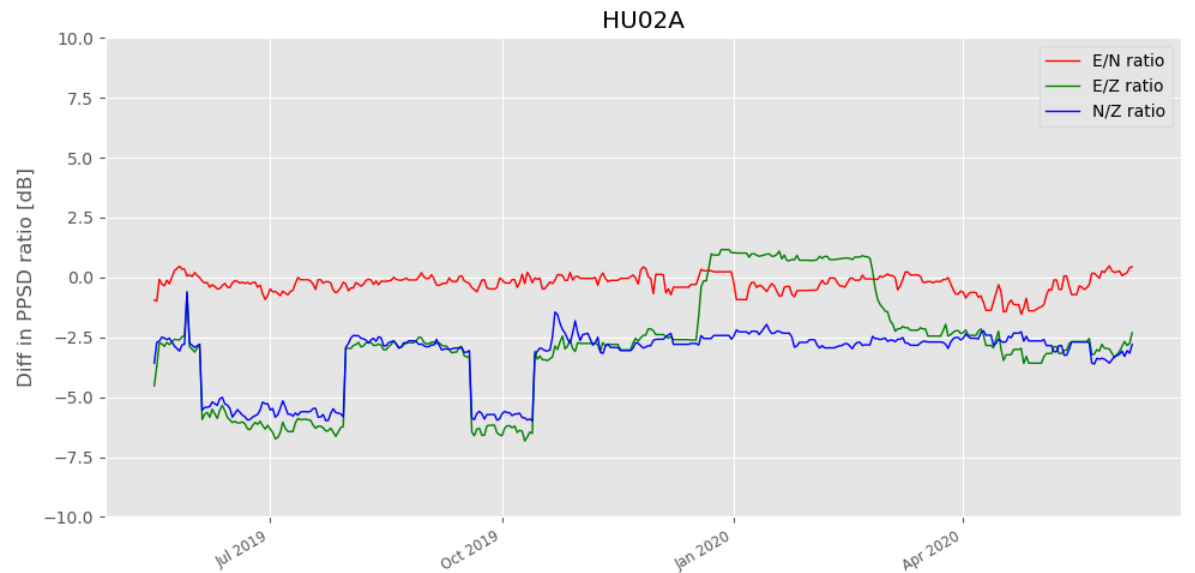
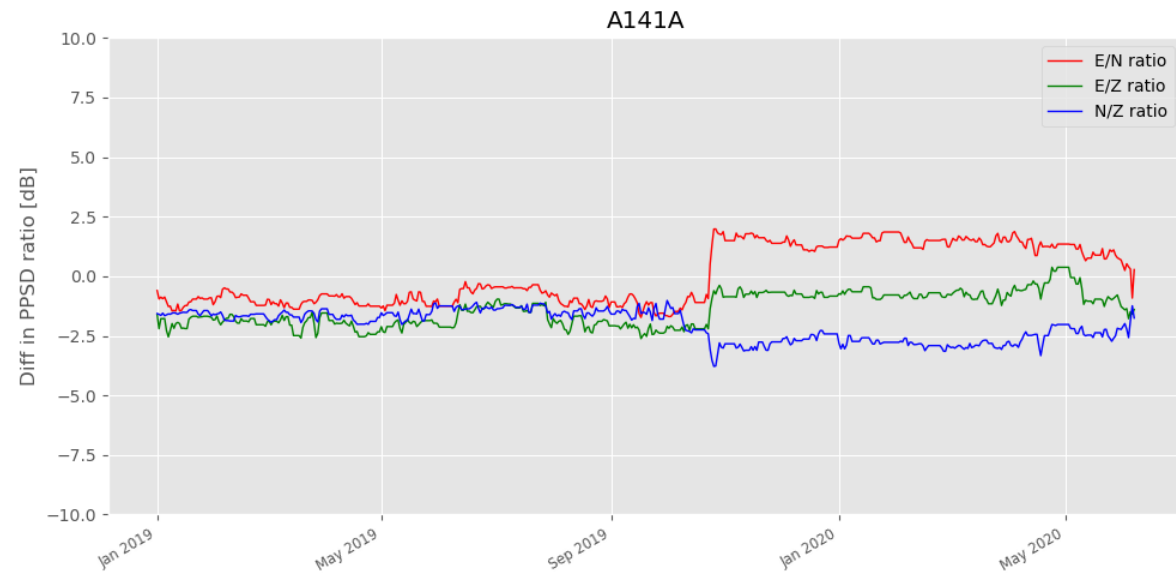
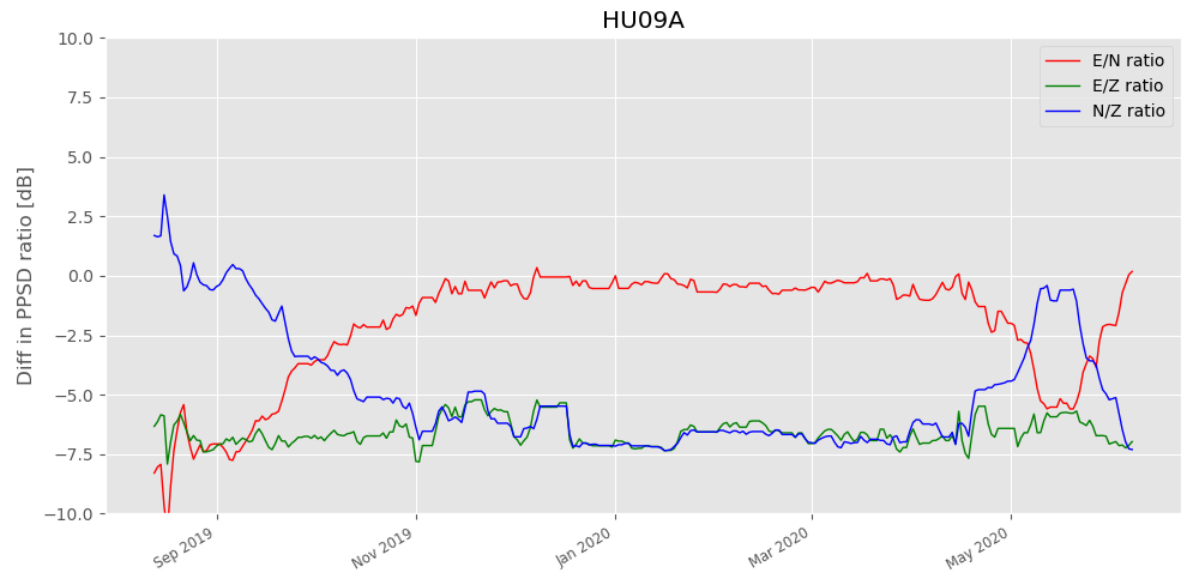
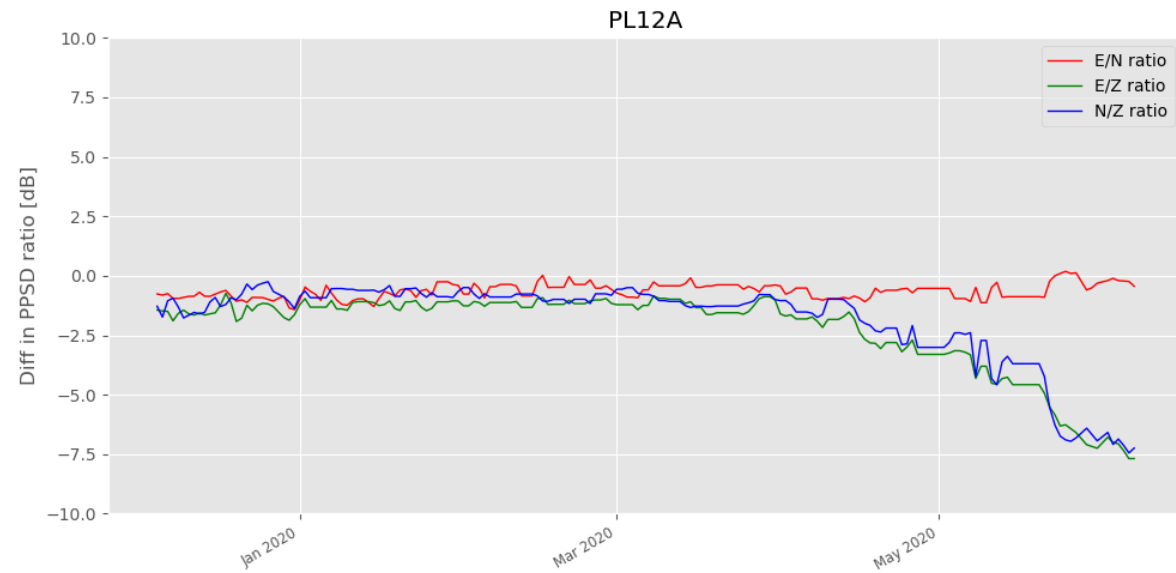


sediments, outliers

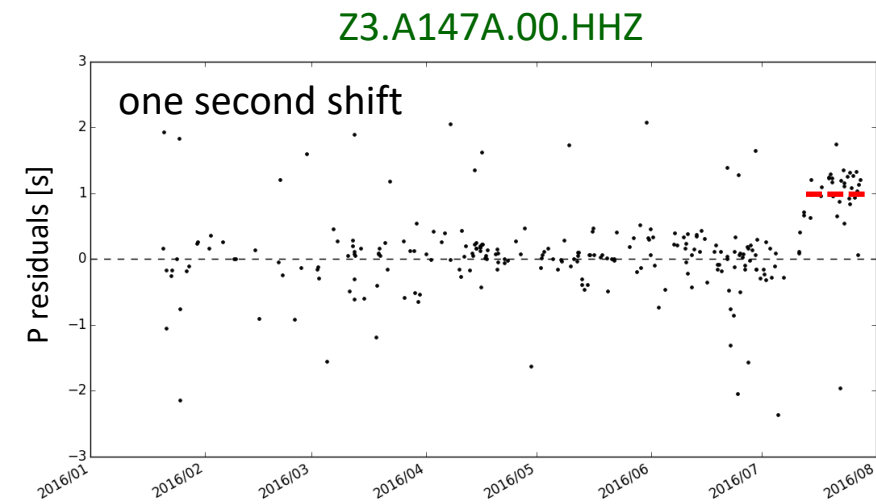
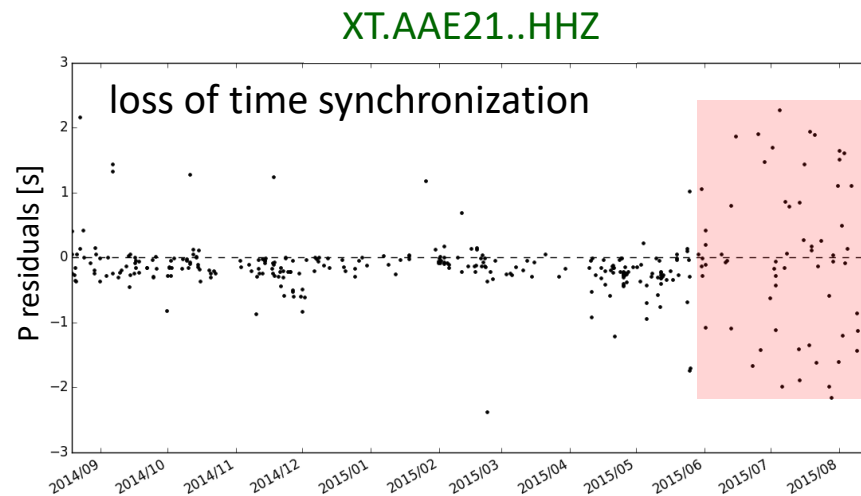
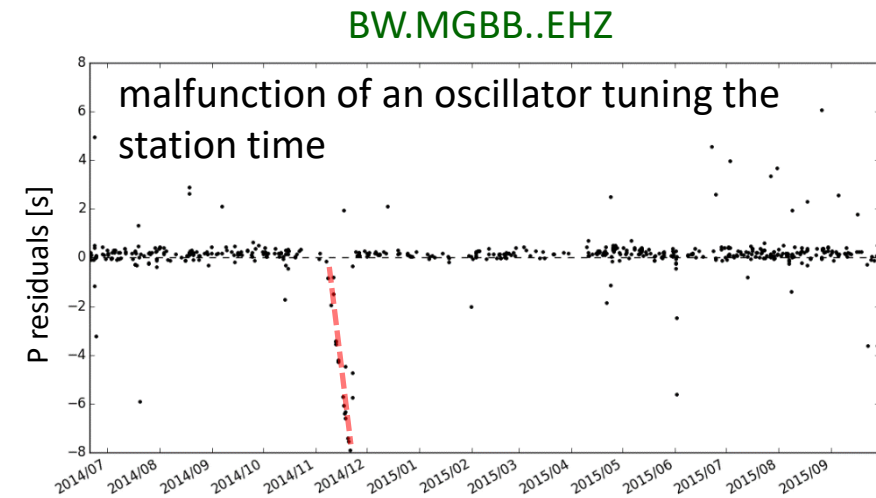
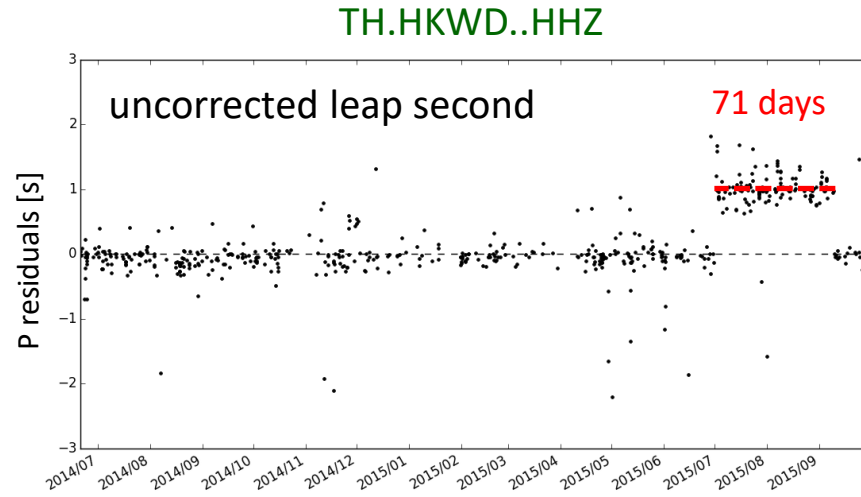


PACASE - amplitude gain issues

- temporal variations of PPSD ratios (for $T = 5s$)
- global trend subtracted



Timing issues - examples



Conclusions



- A non-negligible part of the data and metadata in EIDA is defective or incorrect.
- We can identify a wide group of problems in metadata and data, we have tools to find them. The removal of some defects is simple (by changes in metadata), cooperation with data providers and EIDA is desirable.
- To guarantee high-quality data and metadata from MOBNET stations, we use a double-check for quality control: the hardware control in-situ and the ex-post software data checking
- QC processes are underway to correct this condition
 - ❖ EIDA Issue Tracker - <https://github.com/EIDA/userfeedback/>
 - ❖ User Advisory Group (UAG) - reviews services and suggests improvements to ORFEUS services and products
 - ❖ AdriaArray - Working Group 3 'Data QC'

Thank you for your attention!

Acknowledgment

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Several figures and calculations have been prepared with the use of the Generic Mapping Tools (Wessel and Smith, 1998) and ObsPy (Krischer et al., 2015).

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